

Most of `_ship_hit()` is unchanged. We've moved all the existing code into an `if` block, which tests to make sure the player has at least one ship remaining. If so, we create a new fleet, pause, and move on. If the player has no ships left, we set `game_active` to `False`.

Identifying When Parts of the Game Should Run

We need to identify the parts of the game that should always run and the parts that should run only when the game is active:

```
alien_invasion.py    def run_game(self):
                    """Start the main loop for the game."""
                    while True:
                        self._check_events()

                        if self.game_active:
                            self.ship.update()
                            self._update_bullets()
                            self._update.aliens()

                        self._update_screen()
                        self.clock.tick(60)
```

In the main loop, we always need to call `_check_events()`, even if the game is inactive. For example, we still need to know if the user presses `Q` to quit the game or clicks the button to close the window. We also continue updating the screen so we can make changes to the screen while waiting to see whether the player chooses to start a new game. The rest of the function calls need to happen only when the game is active, because when the game is inactive, we don't need to update the positions of game elements.

Now when you play *Alien Invasion*, the game should freeze when you've used up all your ships.

TRY IT YOURSELF

13-6. Game Over: In *Sideways Shooter*, keep track of the number of times the ship is hit and the number of times an alien is hit by the ship. Decide on an appropriate condition for ending the game, and stop the game when this situation occurs.

Summary

In this chapter, you learned how to add a large number of identical elements to a game by creating a fleet of aliens. You used nested loops to create a grid of elements, and you made a large set of game elements move by calling each element's `update()` method. You learned to control the direction of